

RHDV1, RHDV1a variant and RHDV2

Rabbit haemorrhagic disease virus (RHDV), or calicivirus is a virus that typically causes a rapidly fatal disease in European rabbits (*Oryctolagus cuniculus*). There are three types of RHDV present in Australia (RHDV1, RHDV1a and RHDV2).

The classical RHDV1 strain was first released in Australia in 1996 to control wild rabbits.

Over time the effectiveness of the virus as a biocontrol agent has been decreasing. To boost the RHDV1 strain's effectiveness, state and territory governments around Australia released RHDV1a, a new naturally occurring variant of RHDV1, in March 2017.

RHDV2 is a new strain of the calicivirus that was first reported in Europe in 2010, and was first detected in Australia in 2015. It is not known how the virus entered Australia. It is currently the dominant strain causing deaths in domestic rabbits.

RHDV2 has been found to infect rabbits and hares, but has not been found to infect or kill any native Australian mammals or other introduced species. Both the European brown hare and the European rabbit are invasive species to Australia and considered a threat to Australia's environment and agricultural industry.

RHDV can be spread via direct contact with infected rabbits or their excretions (urine, faeces, saliva), via equipment and clothing, and also by vectors including flies.

To protect pet and production rabbits from possible RHDV infection, owners should implement good hygiene practices, ensure animals are protected through appropriate housing and insect proofing, ensure rabbits are vaccinated, and avoid contact with other rabbits – whether wild (feral) rabbits or pet rabbits from outside the household.

There are now two vaccines available for use in rabbits in Australia. The Cylap® vaccine currently used in Australia is effective against RHDV1 but may only provide partial protection against the RHDV2 strain. The Filavac VHD K C + V vaccine is an inactivated bivalent vaccine for rabbits which can provide protection against both RHDV1 and RHDV2, and this vaccine was recently approved for emergency use in Australia .

Rabbits owners should contact their local veterinarian for advice on rabbit vaccination and biosecurity measures specific to their pet or facility.

More information

Further information about RHDV and recommendations for vaccinating rabbits against all strains of Calicivirus are available on the [Australian Veterinary Association website](https://www.australianveterinaryassociation.org.au/).